UserId

String value

UserId(String)

GameId

String id

GameId(String)

BuildingId

String id

BuildingId(String)

GameSessionId

String value

GameSessionId(String)

ReservationId

String value

ReservationId(String)

TournamentId

String value

TournamentId(String)

TournamentMatchId

String value

TournamentMatchId(String)

TeamId

String value

TeamId(String)

GameType

CHESS
FOOSBALL
DARTS
MONOPOLY
RISK
SLOT_MACHINE
ROULETTE

GameStatus

WAITING
IN_PROGRESS
PAUSED
COMPLETED
ABORTED

GameMachineStatus

AVAILABLE
RESERVED
IN_USE
MAINTENANCE
LOBBY

ReservationStatus

PENDING
CONFIRMED
CANCELLED
EXPIRED

TournamentStatus

DRAFT
OPEN_REGISTRATION
IN_PROGRESS
COMPLETED
CANCELLED

TournamentFormat

SINGLE_ELIMINATION
ROUND_ROBIN

TournamentMatchStatus

SCHEDULED
IN_PROGRESS
COMPLETED
ABANDONED
BYE

WinCondition

WIN
DRAW
ABANDONED
TIMEOUT
TEAM_VICTORY

StopReason

COMPLETED
ABORTED
TIMEOUT

Role

PLAYER
LOCAL_ADMIN
GAME_ADMIN
PLATFORM_ADMIN

Role of(String)
Set~Role~ parse(String)
List~String~ toAuthorityNames(String)

CentralUser

UserId id
String username
String passwordHash
String email
List~String~ roles
Instant createdAt

changePassword(String)
updateRoles(List~String~)

GameDefinition

GameType gameType
String name
int minPlayers
int maxPlayers
boolean teamAllowed
Map~String,Object~ registrationRules
Instant createdAt
Instant updatedAt

Tournament

TournamentId tournamentId
String name
GameType gameType
boolean teamBased
int teamSize
TournamentFormat format
TournamentStatus status
Instant startsAt
Instant endsAt
UserId createdBy
Instant createdAt

Tournament openRegistration()
Tournament cancel()
Tournament startProgress()
Tournament complete(Instant)
Tournament update(String, Instant)

Team

TeamId teamId
TournamentId tournamentId
String name
List~UserId~ members
Instant createdAt

TournamentParticipant

TournamentId tournamentId
String participantId
boolean isTeam
String displayName
Instant registeredAt

TournamentMatch

TournamentMatchId matchId
TournamentId tournamentId
int round
int bracketPosition
String participantA
String participantB
String buildingId
String gameId
String sessionId
String winner
TournamentMatchStatus status
Instant scheduledAt
Instant playedAt
String resultData

TournamentStanding

TournamentId tournamentId
String participantId
int wins
int losses
int points
Integer rank

PlayerStatistics

UserId userId
GameType gameType
int matchesPlayed
int matchesWon
Instant lastPlayedAt

PlayerStatistics mergeIncrement(boolean, Instant)

PlayerMatchFact

String sessionId
UserId userId
BuildingId buildingId
GameType gameType
String tournamentId
boolean won
WinCondition winCondition
Instant endedAt

AggregatedStatistics

String id
BuildingId buildingId
GameType gameType
LocalDate periodStart
LocalDate periodEnd
int totalSessions
int avgDurationSeconds
int totalReservations
int totalAbortedSessions
Map~String,Object~ data

mergeWith(AggregatedStatistics)

RegisteredLocalServer

BuildingId buildingId
String baseUrl
Instant lastSeenAt
boolean isActive

updateLastSeen(Instant)
setActive(boolean)

ProcessedEvent

String eventId
Instant processedAt

ReplicationProgress

String eventId
String serverId

LocalAdminBuilding

UserId userId
BuildingId buildingId
Instant assignedAt

FailedLoginAttempt

String username
Instant attemptedAt
boolean success

LocalUser

UserId userId
String username
String passwordHash
String email
List~String~ roles
Instant eventTime
Instant updatedAt

Game

GameId id
GameType gameType
String name
BuildingId buildingId
GameMachineStatus status
long version

reserve()
startUse()
release()
setMaintenance()
setLobby()
rename(String)

GameSession

GameSessionId id
GameId gameId
GameType gameType
BuildingId buildingId
GameStatus status
Instant startedAt
Instant endedAt
Integer durationSeconds
UserId winnerId
WinCondition winCondition
GameResult result
List~UserId~ participants
Instant pausedAt
int accumulatedPausedSeconds
long version
TournamentMatchId tournamentMatchId
TournamentId tournamentId

complete(GameResult, Instant)
abort(StopReason, Instant)
cancelLobby(Instant)
pause(Instant)
resume(Instant)
calculateDuration()
addParticipant(UserId)
removeParticipant(UserId)

Reservation

ReservationId id
GameId gameId
UserId userId
ReservationStatus status
Instant startTime
Instant endTime
Instant createdAt
long version

canBeCancelled(Clock) boolean
confirm()
cancel()
expire()

TournamentSummaryLocal

TournamentId tournamentId
String name
GameType gameType
boolean teamBased
int teamSize
TournamentStatus status
Instant startsAt
Instant endsAt
List~String~ buildingIds
int participantsCount
boolean deleted
Instant updatedAt

TournamentMatchLocal

TournamentMatchId id
TournamentId tournamentId
int round
int bracketPosition
String participantA
String participantB
GameType gameType
String gameId
TournamentMatchStatus status
Instant scheduledAt

TournamentMatchLocal withStatus(TournamentMatchStatus)

LocalStatistics

GameType gameType
int totalSessions
double avgDuration
int totalReservations
Map~String,Double~ winRateByUser

recalculate(List~GameSession~)

GameDefinitionLocal

GameType gameType
String name
int minPlayers
int maxPlayers
boolean teamAllowed
Map~String,Object~ registrationRules
Instant updatedAt

OutboxEvent

String id
String eventType
String payload
String status
Instant createdAt
Instant sentAt
int retryCount

markAsSent(Instant)
incrementRetry()
markAsFailed()
hasFailed() boolean

OutboxEventStatus

PENDING
SENT
FAILED

RegisteredLocalServerLocal

BuildingId buildingId
String baseUrl
Instant lastSeenAt
boolean active
Instant updatedAt

UserRepository

Optional~User~ findById(UserId)
Optional~User~ findByUsername(String)
save(User)

TournamentRepository

Optional~Tournament~ findById(TournamentId)
save(Tournament)

GameRepository

Optional~Game~ findById(GameId)
List~Game~ findByBuildingId(BuildingId)
save(Game)
deleteById(GameId)

GameSessionRepository

Optional~GameSession~ findById(GameSessionId)
Optional~GameSession~ findActiveByGameId(GameId)
List~GameSession~ findByParticipant(UserId)
save(GameSession)

ReservationRepository

Optional~Reservation~ findById(ReservationId)
Optional~Reservation~ findByUserIdAndGameId(UserId, GameId)
List~Reservation~ findPendingByGameId(GameId)
save(Reservation)

OutboxEventRepository

List~OutboxEvent~ findPending()
save(OutboxEvent)

PushUserToCentralPort

pushUsers(List~UserSyncDto~, String)

PushTournamentSummaryToCentralPort

pushTournamentSummaries(List~TournamentSummaryEventDto~, String)

GameDefinitionLocalRepository

Optional~GameDefinitionLocal~ findByGameType(GameType)
save(GameDefinitionLocal)

LocalAdminBuildingLocalRepository

List~LocalAdminBuilding~ findByUserId(UserId)
save(LocalAdminBuilding)
deleteByUserIdAndBuildingId(UserId, BuildingId)

GameSessionService

GameSession start(GameId, UserId, GameType, BuildingId, List~UserId~, Instant)
GameSession start(GameId, UserId, GameType, BuildingId, List~UserId~, Instant, TournamentMatchId, TournamentId)
pause(GameSessionId, UserId, Instant)
resume(GameSessionId, UserId, Instant)
end(GameSessionId, UserId, GameResult, Instant)
abort(GameSessionId, UserId, StopReason, Instant)

GameStateService

GameStateDto getGameState(GameId)
List~GameStateDto~ getBuildingGames(BuildingId)
List~GameSessionDto~ getActiveSessions(BuildingId)

ReservationService

ReservationDto create(CreateReservationRequestDto)
confirm(ReservationId, UserId)
cancel(ReservationId, UserId)
expirePending(Clock)

StatisticsService

LocalStatistics getStatistics(BuildingId, GameType)
PlayerStatisticsDto getPlayerStatistics(UserId, GameType)
List~GameSessionDto~ getActiveSessions(BuildingId)
BuildingStatisticsDto getBuildingStatistics(BuildingId)

LocalAuthService

LoginResponseDto authenticate(LoginRequestDto)

LocalSignupService

SignupResponseDto register(SignupRequestDto)

UserSyncService

syncUsers(List~UserSyncDto~)

GameCatalogService

GameDto createGame(CreateGameRequestDto)
GameDto updateGame(GameId, UpdateGameRequestDto)
deleteGame(GameId)
List~GameDto~ getGames(BuildingId)

SyncSchedulerService

syncWithCentral()

SessionRecoveryService

recoverSessions()

TournamentLifecycleRequestedService

handleTournamentCreated(TournamentSummaryEventDto)
handleTournamentUpdated(TournamentSummaryEventDto)
handleTournamentDeleted(TournamentSummaryEventDto)

TournamentMatchLocalSyncService

handleMatchScheduled(TournamentMatchScheduledDto)
handleMatchCompleted(TournamentMatchResultDto)

JwtTokenProvider

String generateTokenWithExpiry(User, Instant)
JwtClaims parseToken(String)

JwtAuthenticationFilter InternalApiKeyFilter

LocalAdminBuildingAuthorizationManager CurrentUserService MqttPublisher
publish(String, String, int, boolean)

GameStatePublisher

publishState(GameId, GameMachineStatus)
publishSessionStart(GameSessionDto)
publishSessionEnd(GameSessionDto)

AlertPublisher

publishAlert(String, String)

MqttMessageHandler

handle(String, String)

SessionStartHandler

handle(SessionStartPayload)

SessionPauseHandler

handle(SessionPausePayload)

SessionResumeHandler

handle(SessionResumePayload)

SessionEndHandler

handle(SessionEndPayload)

HeartbeatHandler

handle(HeartbeatPayload)

CentralSystemRestAdapter

syncUsers(List~UserSyncDto~)
syncTournamentSummaries(List~TournamentSummaryEventDto~)
syncGameDefinitions(List~GameDefinitionEventDto~)

RegisterLocalServerAdapter

register(BuildingId, String)

GameResult

TournamentSummaryLocalRepository TournamentMatchLocalRepository

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